Huibin (Matt) Li

PhD Student · CVISS Lab · University of Waterloo

h844li@uwaterloo.ca matt-v50.github.io Matt-V50 huibin-li

Research Interests

3D computer vision, neural rendering, Gaussian Splatting, computer graphics, 3D generative models.

Education

University of Waterloo Waterloo, Canada Sep 2024 – Present
Ph.D. in Structures, Mechanics & Construction Engineering
- Advisor: Prof. Chul Min Yeum, CVISS Lab
- Research: 3D Gaussian Splatting for scene reconstruction and understanding

University of Chinese Academy of Sciences (UCAS) Beijing, China Sep 2021 – Jun 2024
M.Eng. in Electronic and Information Engineering (Artificial Intelligence) GPA 3.87/4
- Advisor: Prof. Jianwei Guo, Institute of Automation, Chinese Academy of Sciences
- Thesis: High-quality automatic texture reconstruction and intelligent generation for 3D digital content production

Yancheng Institute of Technology Jiangsu, China Sep 2011 – May 2015
B.S. in Computer Science Top 10%

Publications

- **Huibin Li**, Chul Min Yeum. **Warp-2DGS: Learnable Warped Kernels for 2D Gaussian Splatting.** *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. **Spotlight.**
  [Project page] [Code]
- Pu Li, Jianwei Guo, **Huibin Li**, Bedrich Benes, Dong-Ming Yan. **SfmCAD: Unsupervised CAD Reconstruction by Learning Sketch-based Feature Modeling Operations.** *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
  [Code]

Patent
- Monthly active user prediction method based on LSTM neural network, storage medium and computer equipment. CN112116381A.

Research Experience

Graduate Research Assistant, CVISS Lab, University of Waterloo Sep 2024 – Present
- Learnable deformable kernels for 2D Gaussian Splatting that suppress Gibbs artifacts at sharp boundaries (NeurIPS 2026 Spotlight)
- Floater removal for 3D Gaussian Splatting with attention-based transformers
- Interactive oriented bounding box extraction from 3D Gaussian Splatting for scene understanding

Graduate Research Assistant, Institute of Automation, Chinese Academy of Sciences Sep 2021 – Jun 2024
- UV texture reconstruction for 3D meshes with deep learning, improving LPIPS and PSNR over baselines
- Diffusion-based text-guided texture generation for 3D assets
- Neural rendering pipelines for 3D scene reconstruction (PyTorch, CUDA)

Industry Experience

Research Intern, AI Lab, Well-Link Times Beijing, China Nov 2023 – Aug 2024

- Applied diffusion models to generate high-quality texture maps for 3D models, improving LPIPS and PSNR
- Built interactive Blender add-ons addressing spatial consistency in 3D modeling workflows
- Reduced game asset creation time from days to minutes through automated generation
- Developed an LLM-based prompt enhancement system, cutting average prompt length from 75 to 5 words

Algorithm Engineer, Tingyun Beijing, China Apr 2015 – Jul 2021

- Built LSTM-based time-series forecasting of monthly active users for capacity planning
- Designed an entropy-based URL aggregation algorithm, reducing storage requirements by 95%
- Developed real-time anomaly detection for high-dimensional application performance metrics

Teaching

Teaching Assistant, CIVE 221, University of Waterloo Fall 2026

Talks

- *The development and future of large models*, invited talk, China University of Political Science and Law, Dec 2023
- *Popular science on large models*, invited talk, Youth Science and Technology Innovation Salon, UCAS, Jan 2024

Skills

Programming Python, C++, CUDA, JavaScript/TypeScript, Java, SQL
Tools PyTorch, Blender, Git, Docker, Linux, LaTeX